University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-4 Odd Semester Examination-2025
Course: CSE4141 (Microcontroller Interfacing)
Marks: 52.50 Time: 3:00 Hours

[N.B. Answer any Six (06) questions taking Three (03) from each section]

Section-A

1. (a) Define computer interfacing with an example. 2
   (b) Explain why serial data transfer is mostly preferred over parallel data transfer. Give reasons. 2
   (c) Distinguish between absolute and partial address decoding with example. 3
   (d) What is DMA? What are the uses of DMA? 1.75

2. (a) Explain how device select pulse is generated to select a device prior to data transfer. 2.00
   (b) Discuss how a processor is interfaced with a common anode seven-segment LED using a decoder and a latch as an output port. Also write the program to display digit 7 to the output port. 3.75
   (c) Discuss an interfacing circuit to design a safety control system for home appliances using a decoder, a buffer as an input port and a latch as an output port. Also write the program to operate the controlling system. 3.00

3. (a) In peripheral interfacing is it possible to write the instructions LDA 8000H and then STA 8000H. Explain your answer. 2.00
   (b) Discuss how time delay is calculated using loop within loop technique for generating certain delay in between two consecutive events. 3.75
   (c) Write a program to count from 0 to 9 (modulo 10) with a one second delay between each count. The output will be shown in a port with address 01H and at the end of each count the sequence will be repeated. Assume the clock frequency of MPU is 1MHz. 3.00

4. (a) What do you mean by Programmable Peripheral Interfacing (PPI)? 1.00
   (b) Design a square-wave generator with a pulse width of 100μs by using the 8155 timer. Set up the timer in Mode 1 if the clock frequency is 3 MHz. 4.00
   (c) Interface 8255A ports in mode 0 with processor for the following conditions: 3.75
   (i) Use port B and port C_L as input ports to read inputs from DIP switches and port A and port C_U as output ports to display output at leds.
   (ii) Write the program to read the DIP switches and display the reading from port B at port A and from port C_L at port C_U.

Section-B

5. (a) What is Cyclic Redundancy Check (CRC)? What are the application of CRC? 2.75
   (b) A sender wants to transmit the data bit stream `1101011011`. The system uses the generator 6

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polynomial $G(x) = x^3 + x^2 + 1$ for a CRC.

i) Find the binary divisor corresponding to the generator polynomial.

ii) Calculate the CRC bits (the remainder) that will be appended to the data.

iii) Determine the final codeword (the actual bit string sent to the receiver).

iv) Assume the receiver gets the exact transmitted codeword. Show how the receiver verifies that the data is error-free.

6. (a) Compare I^2C with SPI and USART. Where is I^2C used? 3

(b) I^2C that allows multiple chips to talk to each other on the same circuit board. Discuss the basic architecture of I^2C with appropriate block diagram. 4

(c) What are the limitations of I^2C? 1.75

7. (a) Write the features of SPI. 1

(b) Consider a microcontroller acting as an SPI Master connected to a single SPI Slave (such as a digital sensor) using the standard 4-wire interface: `SCLK`, `MOSI`, `MISO`, and `SS` (Slave Select). 7.75

i) Explain briefly how full-duplex communication takes place simultaneously on the `MOSI` and `MISO` lines during each clock cycle of the `SCLK` signal.

ii) The Master wants to send an 8-bit command byte `0xA5` to the Slave to request data, while the Slave simultaneously transmits an 8-bit sensor reading `0x3C` back to the Master. iii) Illustrate the contents of the internal shift registers in both the Master and the Slave after all 8 clock pulses have completed. State the final received data values in both the Master's receive buffer and the Slave's receive buffer assuming a standard shift-out order (MSB first).

8. (a) Suppose you want to connect three microcontrollers for data exchange. Which protocol (UART, SPI, or I^2C) would you choose and why? Describe the protocol with proper justification. 4.00

(b) Suppose, two microcontrollers use the SPI communication protocol. If the clock frequency is 4 MHz, how long will it take to transmit 32 bits of data? 2.75

(c) Explain why pull-up resistors are needed in I^2C lines. 2.00

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part IV, Odd Semester Examination-2025
Course Code: CSE-4191, Course Title: Blockchain
Time: 3 Hours Full Marks: 52.5

Answer any Six (06) questions taking at least Three from each of the sections.

Section-A

1. a) What is a block in a blockchain? Describe the major components of a block with a suitable diagram. Explain how blocks are linked together to form a blockchain, and illustrate the process with an example. 5.00

   b) Explain the concept of immutability in blockchain. Explain how the hash chain ensures that data cannot be changed retroactively. 3.75

2. a) What is digital signature? A blockchain-based university certificate system allows students to receive digitally signed certificates. A student can present the certificate to an employer, who can verify its authenticity using the issuer's public key. 5.00
   Questions:
   i) Why is a digital signature required in this system?
   ii) How can the employer verify that the certificate was issued by the university?
   iii) What happens if someone modifies the student's grade on the certificate?
   iv) Identify the security properties provided by the digital signature.

   b) A block contains eight transactions: **T1, T2, T3, T4, T5, T6, T7, and T8**. Their hashes are used to construct a Merkle tree. If a node wants to verify only **T5**, does it need to download all eight transactions? Explain how a Merkle proof can be used to verify **T5**. 3.75

3. a) What do you mean by consensus in blockchain technology? Differentiate between Proof of Work and Proof of Stake based on their block-validation processes. 3.00

   b) What is the Byzantine Army Problem? How can the presence of traitorous generals make consensus difficult in the Byzantine Army Problem? Give a real life example. 3.00

   c) Define Byzantine fault. What are the three sub protocols of Practical BFT algorithms. Explain them. 2.75

4. a) What is Bitcoin? How does Bitcoin differ from traditional currency in terms of control, transaction verification, and record keeping? 3.00

   b) What is a Bitcoin hardware wallet? Describe its purpose and explain, step by step, how it can be used to receive and send Bitcoin. Use a suitable example to illustrate both processes. 5.75

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Section-B

5.	a)	"Smart contracts operate on-chain but often depend on off-chain resources." Explain this statement with suitable examples.	3.00
	b)	A Bangladeshi buyer and a German seller enter into a smart contract. A dispute arises regarding the performance of the contract. Analyze the legal issues that may arise if the contract does not specify the governing law and venue.	2.00
	c)	What is oracle? Classify different types of inbound oracles.	3.75
6.	a)	What is an Ethereum account? What are the properties EOA and CA?	3.75
	b)	A contract account may have ETH balance. Explain uses of ETH balance with a complete example of crowdfunding through CA.	3.00
	c)	What is Gas in Ethereum blockchain? Can the Contract use its ETH balance to Pay Gas? Why?	2.00
7.	a)	What is an Ethereum transaction? What are the major purposes for which an Ethereum transaction can be used?	2.00
	b)	Explain the major steps performed by the Ethereum network after receiving a transaction. Alice wants to send 1 ETH to Bob. Her transaction has a nonce of 5 and a gas limit of 21,000. Describe the major steps the Ethereum network will perform before Bob receives the ETH.	2.75
	c)	What are message call and contract creation transactions in Ethereum blockchain? Explain them with examples.	4.00
8.	a)	What is a decentralized application (DApp)? List the three architectural layers of a DApp.	3.00
	b)	Describe the truffle development workflow. What are the roles of Truffle, Ganache and Metamask in this workflow?	3.00
	c)	Write a simple JavaScript code snippet (using web3.js) that connects to a deployed smart contract and calls a function to get the stored message.	2.75

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. (Engg.) Part-4 Odd Semester Examination-2025
Course: CSE4121 (Object Oriented Design and Design Patterns)
Marks: 35 Time: 2:00 Hours

[N.B. Answer any Four (04) questions taking Two (02) from each section]

Section A

1. a) Differentiate between Conceptual Design and Technical Design with suitable examples from the Dhaka Urban Management System (DUMS). 3

b) Agricultural Automation System (AAS) is a smart farming platform that uses IoT sensors, AI models, drones, and automated machines to support farmers. The system monitors soil conditions, controls irrigation, detects pests, predicts harvesting time, and provides recommendations through a mobile application. Now based on the above scenario: 4.75

① Identify at least four important objects from the problem domain and classify each object as an Entity Object, Control Object, or Boundary Object.

② Develop Class–Responsibility–Collaborator (CRC) cards for the following classes: Farmer, SensorModule, IrrigationController, and WeatherService. Each CRC card should include the Class Name, Responsibilities, and Collaborators.

c) An Agricultural Automation System requires automatic irrigation using IoT sensors. Farmers want a low-cost system, but engineers recommend expensive, high-accuracy sensors for better prediction. Based on the above scenario: Analyze the requirement conflict and explain how developers can reach a compromise among cost, accuracy, and system performance. 1

2. a) A team argues that UML models are unnecessary because modern IDEs can automatically generate code. Evaluate this argument and justify the importance of UML in software development. 1.75

b) A Dhaka Urban Management System contains different components such as citizens, garbage bins, waste collection trucks, collection schedules, and monitoring applications. Proper modeling of relationships among these objects is essential for effective software design: 4

Analyze the relationships among these objects and:

① Determine where association, aggregation, and composition relationships should be used.

② Justify your choice based on object dependency and lifetime.

③ Draw UML diagrams representing each selected relationship.

c) A smart farming company wants to develop an Agricultural Automation System that manages different types of machines, including irrigation systems, harvesters, and 3

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pesticide spraying machines. A developer proposes creating separate classes for each machine without using inheritance. Analyze this design decision and:

① Explain the limitations of creating independent classes.
② Redesign the system using a suitable parent-child class hierarchy.
③ Explain how the system can achieve improved reusability

3. a) In a food delivery application, an order passes through several stages: **Placed → Preparing → Out for Delivery → Delivered.** The behaviour of the order object changes depending on its current stage — for example, an order can be cancelled while it is *Placed*, but not once it is *Out for Delivery*. 4.5

Which design pattern is most suitable for implementing this stage-dependent behaviour?

Explain it with a UML class diagram.

b) Suppose you are designing a refund approval system for an e-commerce platform with the following authorization scenario:

① For refund amount Tk. < 1,000, only the support agent's approval is sufficient.
② For refund amount Tk. 1,000 to 50,000, approval from the customer service manager is needed. 3
③ For refund amount Tk. > 50,000, approval from the operations director is needed.

Implement this behaviour using an appropriate software design pattern.

Section B

4. a) Explain the purpose of the Gang of Four (GoF) design pattern catalog. 2

b) Suppose an application requires exactly one shared configuration object throughout the system. Which design pattern would you use? Explain how you would apply it. 3

c) How do structural design patterns contribute to the organization of software? Explain with an example. 3.75

5. a) The Dhaka Urban Management System receives thousands of citizen complaints every day regarding waste collection, traffic problems, water supply, and emergency services. When the status of a complaint changes, different users should automatically receive updates: 4

① Citizen receives complaint status notification.
② Department officer receives update information.

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③ City administrator receives monitoring alerts.

④ The complaint management system should notify all registered users automatically whenever the complaint status changes.

According to the above scenario and requirements, implement this behavior using an appropriate software design pattern.

b) A university has several levels of approval: Teacher → Chairman → Dean → Vice Chancellor. Apply the Chain of Responsibility Pattern to this approval system 2.75

c) A Smart Agricultural Automation System uses an existing irrigation controller that accepts data from a standard soil moisture sensor. A newly purchased IoT soil sensor provides moisture information using a different interface. The existing irrigation controller cannot directly communicate with this new sensor. According to above scenario answer the following Question: 2

① What problem does the Adapter Design Pattern solve?

② Explain how the Adapter Pattern can integrate the new IoT soil sensor without modifying the existing irrigation controller.

6. a) Explain the responsibilities of the Model, View, and Controller in the MVC Pattern. 2

b) A method contains 100 lines of code and performs validation, database access, calculations, and report generation. Identify the likely code smell and suggest an appropriate refactoring approach. 3

c) Create a refactored design for a system suffering from a Large Class and Duplicated Code. Identify how responsibilities should be redistributed among multiple classes. 3.75

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University of Rajshahi
Department of Computer Science and Engineering
B.Sc. Engineering Part IV, Odd Semester Exam-2025
Course Code: CSE-4111 Course Title: Parallel and Distributed Systems

Time: 3 Hours Full Marks: 52.5

Answer any Six (06) questions taking at least Three from each of the sections.

Section-A

1.(a) Define parallel processing and explain how it differs from distributed processing. 3.75

(b) Explain the major types of parallelism with suitable examples. 3.00

(c) Discuss the major limitations of parallel computing. 2.00

2.(a) Explain the GPU memory hierarchy with a suitable architectural diagram. Compare registers, shared memory, L1 cache, L2 cache, constant memory, and global memory in terms of location, access speed, and scope. 4.75

(b) Consider a two-dimensional array A[N][N]. Compare the cache behavior of the following two traversal methods: 4.00

```
for i = 0 to N-1                    and    for j = 0 to N-1
  for j = 0 to N-1                           for i = 0 to N-1
    access A[i][j]                             access A[i][j]
```

Explain which is more cache-friendly and why.

3.(a) Write an MPI program to calculate the sum of N integers using collective communication. Explain the role of MPI_Bcast, MPI_Scatter, and MPI_Reduce. 5.00

(b) What is Cache coherence protocol? What are the characteristics of directory based Cache coherence protocols? 3.75

4.(a) *'Data dependency is a situation in which a program statements refer to the data of a preceding statement'*. What are the types of it? Explain with examples. 4.00

(b) Consider the following code fragment of a program— 2.75

S1: Load R1,A
S2: Add R2,R1
S3: Move R1,R3
S4: Store B,R1

Find the dependencies between the statements in the code fragment.

(c) Explain how *I2* is resource dependent on *I1* in the fragment given below— 2.00

I1: A=B+C
I2: D=G+H

Hence, draw the dependency graph.

Section-B

5.(a) What is First–Second Transposition Sort? Explain its basic working principle and how it sorts an unsorted sequence. 5.00

(b) Explain OpenMP loop scheduling. Compare static, dynamic and guided scheduling with suitable examples. 3.75

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6.(a) Consider two processes *P1* and *P2*. Their inputs are *I1* and *I2*, and outputs are *O1* and *O2*. What are the Bernstein's conditions to be satisfied to execute them in parallel? Give the properties of parallelism between them and other statements. 4.00

(b) Consider the following processes *P1, P2, P3, P4* and *P5* are in program order. 4.75

*P1: C=D*E;*

P2: M=G+C

P3: A=B+C

P4: C=L+M

P5: F=G/C

Find the pair of statements executable in parallel satisfying the Bernstein's conditions.

7.(a) What is CUDA? Write a CUDA program to perform vector addition. 5.00

(b) What is array indexing in CUDA? Explain how blocks and threads are used to assign individual array elements to CUDA threads. A CUDA kernel uses blockDim.x = 128. Determine the global thread index for: 3.75

blockIdx.x = 0, threadIdx.x = 10
blockIdx.x = 2, threadIdx.x = 20
blockIdx.x = 5, threadIdx.x = 50

8.(a) What is a dependency graph? Explain its role in identifying parallelism in a program. 2.75

(b) What is the difference between a dependency graph and a task graph? Explain with a suitable example. 3.00

(c) Given the operations: 3.00

```
L.D      F0, 0 (R1)
ADD.D    F4, F0, F2
S.D      F4, 0(R1)
L.D      F0, -8(R1)
ADD.D    F4, F0, F2
S.D      F4, -8(R1)
```

Draw the dependency graph and identify all operations that can be executed simultaneously.

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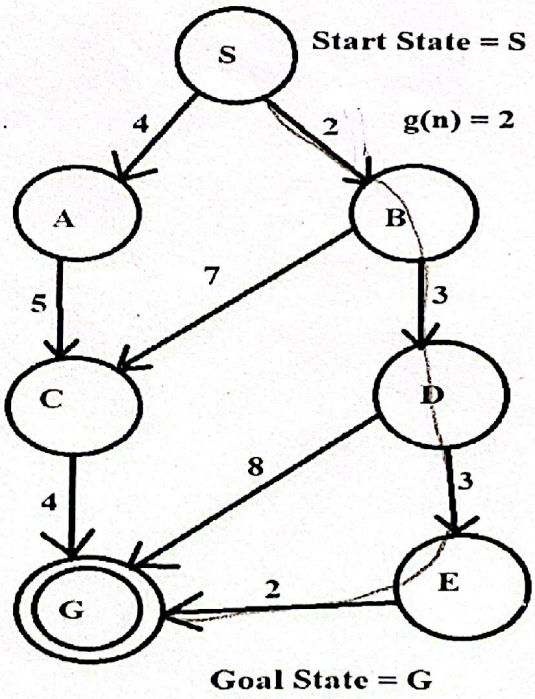
Department of Computer Science and Engineering
B.Sc. Engineering Part-IV, Odd Semester, Examination–2025
Course: CSE-4131 (Artificial Intelligence)

Time: 3 Hours **Full Marks: 52.5**

[N. B. Answer any Six questions taking at least Three questions from each section.]

SECTION-A

1	(a)	Explain the following terms: (i) Artificial Narrow Intelligence (ANI) (ii) Artificial General Intelligence (AGI). Give examples.	3.00
	(b)	Differentiate between declarative, procedural, and heuristic knowledge.	2.00
	(c)	Discuss the advantages, limitations, and future possibilities of Artificial Intelligence.	3.75
2	(a)	Explain basic types of agents in real world environment. What are the factors depending on Rationality of an agent?	3.00
	(b)	What are the basic structures of intelligent agents? Describe the structure of Learning based agents?	3.75
	(c)	Explain Turing test environment for artificial agents.	2.00
3	(a)	Discuss informed search algorithms for AI. Write the applications of informed search algorithms.	2.25
	(b)	How does A* search differ from Greedy Best-First Search?	2.50
	(c)	Consider the following Figure. Find out the goal path using A* search algorithm.	4.00



State	h(n) Value
S	4
A	3
B	5
C	4
D	5
E	2
G	0

4	(a)	Define Inference rules with example. Why is resolution required in AI?	2.50
	(b)	Describe the basic steps in Resolution.	2.50

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(c) Consider the following facts: 3.75

i) Alice is a student.
ii) All students study.
iii) Everyone who studies works hard.
iv) Everyone who works hard passes the examination.

GOAL: Does Alice pass the examination?

Prove the goal by resolution.

SECTION-B

5. (a) Discuss the role of a basic neuron in a neural network. Why are hidden layers needed in a neural network? 2.75

(b) Explain weights and biases in a neural network. Why are they important? 3.00

(c) What happens if the learning rate is too high or too low? Why can a neural network perform well on training data but poorly on unseen data? 3.00

6. (a) Explain the basic components of Convolution Neural Network (CNN). 2.50

(b) Discuss the application of fine tuning in Convolution Neural Network (CNN). 2.50

(c) Define Transfer Learning. Describe common steps for Transfer Learning with examples. 3.75

7. (a) Discuss Deep Neural Network (DNN). Why are multiple hidden layers used in a DNN? Explain an epoch, batch size, and iteration during DNN training. 3.00

(b) Explain why nonlinear activation functions are important in deep learning. Why is normalization of input data important for training deep neural networks? 3.00

(c) Why can two DNNs trained on the same dataset produce different results? What is the purpose of a loss function in a DNN? 2.75

8. (a) Describe monotonic and non- monotonic system with examples. 2.75

(b) What is the role of the knowledge acquisition process, and why is it often difficult? 3.00

(c) Explain reinforcement learning algorithm with diagram. 3.00

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